

Library Management System - Project Report

Class: BCS-4D

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1. Project Overview

The Library Management System (LMS) is a web-based application designed to automate and simplify the operations of a library. It eliminates manual record-keeping and provides a centralized digital system for managing books, members, and transactions. The system is built as a Single Page Application (SPA), ensuring a smooth and responsive user experience without page reloads.

The main objective of this project is to improve efficiency in library operations such as book issuance, returns, fine calculation, and inventory management. It also provides real-time data tracking and reporting features for better decision-making.

2. Technology Stack

The project is developed using a combination of modern web technologies. The frontend is built using HTML5, CSS3, and Vanilla JavaScript. JavaScript Fetch API is used to communicate with the backend without refreshing pages, enabling SPA behavior.

The backend is developed in PHP using a REST-style API structure. It processes all requests through a centralized controller and returns JSON responses to the frontend. This makes the system modular and scalable.

The database used is SQLite, accessed through PHP Data Objects (PDO). SQLite is lightweight and does not require a separate server, making it ideal for small to medium-scale applications and academic projects.

3. Database Architecture

The database follows a relational structure with multiple interconnected tables. The main tables include members, books, categories, issued_books, and fines. Each table is designed with proper relationships to maintain data integrity.

Advanced SQL views are used to simplify complex queries and reporting. These views handle tasks such as identifying overdue books, calculating fines, and generating dashboard statistics. This reduces backend complexity and improves performance.

4. Core Features

The system provides full CRUD (Create, Read, Update, Delete) functionality for both books and members. Librarians can easily add new books, update existing records, or remove outdated entries.

The issuing and returning module ensures proper tracking of book circulation. When a book is issued, the available quantity decreases automatically and a due date is assigned. Upon return, the system checks for overdue status and calculates fines if necessary.

The fine management system automatically calculates penalties based on overdue days. A fixed rate per day is applied, and records are stored for transparency and tracking of unpaid dues.

5. System Architecture

The application follows a Single Page Application architecture. The frontend dynamically updates content using JavaScript without page reloads. All requests are sent to a central API file using action-based routing.

The backend is structured in a modular way, separating database logic, API handling, and initialization scripts. A singleton database connection ensures efficient resource usage.

6. User Interface Design

The UI is designed with a modern dark theme using elegant gold accents, giving it a premium library-like appearance. It is fully responsive and works across different screen sizes including desktops and tablets.

Interactive components such as modals, forms, and toast notifications improve user experience by providing instant feedback without navigating away from the current view.

7. Conclusion

The Library Management System successfully demonstrates the implementation of a full-stack web application using PHP and JavaScript. It automates traditional library processes

and introduces efficiency through digital management. The system is scalable, maintainable, and can be further extended into a cloud-based solution in the future.